- *Autonomous agent*—As we've seen in previous chapters, agents with the ability to plan can complete a series of complex tasks for various goals. Planning will be an essential element of any autonomous agentic system.
- *Collaborative workflows*—Think of these as agents or assistants that sit alongside coders or writers. While these workflows are still in early development, think of a workflow where agents are automatically tasked with writing and executing test code alongside developers. Planning will be an essential part of executing these complex future workflows.
- *Game AI*—While applying LLMs to games is still in early stages, it isn't hard to imagine in-game agents or assistants that can assist or challenge the player. Giving these agents the ability to plan and execute complex workflows could disrupt how and with whom we play games.
- *Research*—Similar to collaborative workflows, these agents will be responsible for deriving new ideas from existing sources of information. Finding that information will likely be facilitated through extensive tool use, which will benefit from coordination of planning.

As you can see, planning is an essential part of many LLM applications, whether through coordination of tool use or otherwise. In the next section, we look at the next component of reasoning and how it can be applied to the same application stack.

11.5.2 Application of assistant/agentic reasoning

Reasoning, while often strongly associated with planning and task completion, is a component that can also stand by itself. As LLMs mature and get smarter, reasoning is often included within the LLM itself. However, not all applications may benefit from extensive reasoning, as it often introduces a thinking cycle within the LLM response. Table 11.2 describes at a high level how the reasoning component can be integrated with various LLM application types.